

William R. Borrelli

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SUMMARY

Computational scientist with expertise at the intersection of atomistic simulation, data science/machine-learning, and statistical mechanics and thermodynamics. Seeking industry roles that involve quantitatively rigorous decision-making and creating actionable data-driven insights.

PROFESSIONAL EXPERIENCE

Graduate Researcher - Department of Chemistry & Biochemistry

May 2022 – Present

University of California, Los Angeles

Los Angeles, CA

- Devised and executed computational research on the solution phase chemistry of hydrated electrons and alkali dimers
- Managed and collaborated on the research projects of undergraduate students, junior graduate students, and an experimentally trained postdoctoral researcher
- Published numerous peer-reviewed publications in prestigious chemistry and chemical physics journals

Computational Materials Science Intern

June 2025 – August 2025

Corporate Research in Materials Engineering at Robert Bosch LLC

Watertown, MA

- Developed and bench-marked a computational workflow for calculating thermodynamic measures of solution non-ideality for corrosion modeling
- Collaborated with an interdisciplinary and international team of materials scientists, engineers, and physicists to guide industry research and development goals

ACCESS Allocation Manager

September 2024 - Present

University of California, Los Angeles

Los Angeles, CA

- Wrote peer-reviewed proposals resulting in grants of over 1 million high-performance computing CPU hours for computational chemical physics research

Undergraduate Researcher

October 2019 – July 2021

Fordham University

The Bronx, NY

- Wrote Python code for automated sampling of high-dimensional chemical space constrained by attainable reaction conditions

EDUCATION

University of California, Los Angeles

Los Angeles, CA

PhD in Theoretical & Computational Chemistry

September 2021 – 2026 (Expected)

Fordham University (Magna Cum Laude)

The Bronx, NY

B.S. in Chemistry, minor in Psychology

August 2017 – May 2021

SELECTED PUBLICATIONS

(9 peer-reviewed manuscripts with 6 as first or co-first author)

- Roles of H-Bonding and Hydride Solvation in the Reaction of Hydrated (Di)electrons with Water to Create H₂ and OH⁻ (J. Chem. Theory Comput. 2024, 20, 16, 7337–7346)
- Using Machine Learning to Understand the Causes of Quantum Decoherence in Solution-Phase Bond Breaking Reactions (J. Phys. Chem. Lett. 2024, 15, 4, 903–911)

PRESENTATIONS & OTHER PROFESSIONAL INVOLVEMENT

SoCal Theochem 2023 - Talk

October 14, 2023

University of California, Los Angeles

Los Angeles, CA

- Title: Solvation Entropy Calculations for Different Models of the Hydrated Electron

UCLA Chemistry & Biochemistry Graduate Admissions Student Reviewer

December 2024

University of California, Los Angeles

Los Angeles, CA

- Collaborated with faculty members in evaluating PhD student applications for the Theory and Computation Specialization at UCLA

Co-organizer of SoCal Theochem 6 Conference

October 2023

University of California, Los Angeles

Los Angeles, CA

- Built and maintained the conference website coded in HTML - allowed for hyperlink registration and an embedded map of UCLA and the surrounding area